Readout-compatible CA1 plasticity supports cue-dependent reconfiguration and partial-cue recall in a minimal hippocampal model

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Abstract

Rapid episodic memory requires more than storing informative content: retrieved content must be expressed in a form that downstream circuits can interpret. We examine this constraint in a minimal hippocampal model in which entorhinal cortex (EC) generates a CA3-like retrieval key and an instructed CA1 target, plastic CA3–CA1 weights associate the key with that target, and a fixed pretrained CA1–EC decoder reconstructs the input. Pretraining establishes a representational basis before rapid episodic learning, allowing the model to test whether a new CA1 state remains compatible with an existing readout. Across 20 paired simulations, aligned CA1 instructions supported one-shot reconstruction (mean output–target cosine 0.804), whereas a fixed permutation of the instructed coordinates impaired reconstruction (0.136; paired difference 0.668, 95% confidence interval 0.648–0.688). Applying the corresponding permutation to the decoder restored performance (0.804), showing that the target retained its information while becoming unreadable to the unchanged decoder. On a cue-guided circular track, exchanging cue identities produced heterogeneous CA1 tuning, including spatially stable, rate-like, and field-shifting responses. During frozen recall, selective degradation of lateral or medial EC inputs produced complementary impairments: lateral EC degradation preferentially reduced cue decoding, whereas medial EC degradation reduced position decoding while sparing cue identity. A dense common-key control failed to distinguish cue or position states. These simulations support a limited circuit-level proposal: CA3-like keys can select stored associative content, while rapid CA1 plasticity reformats that content within a structured representational basis that remains legible to an established output pathway. The model does not establish long-term code stability or recurrent CA3 pattern completion.

1 Introduction

The role of the hippocampus is commonly viewed as a fast-learning system able of binding together elements of current experiences and recall them upon partial input. Indexing of memory traces and learning system models formalize this role as rapid episodic consolidation that complement slower plasticity timescales in the neocortex [14, 20]. Within these perspectives, an efficient memory system should be able to distinguish the current episode from related experiences, as well as retrieve task-relevant content through pattern completion while maintaing a form that downstream circuits can parse. The first two points have been extensively addressed by hippocampal memory theory. Here we place focus on the third, investigating the preservation of output decodability with special emphasis on the hippocampal subfield CA1. An important motivation is that of going past an intepretation of CA1 as merely a relay of episodic traces but rather an adaptive representational interface through which successive experiences can be organized within a structured and sufficiently stable neural space.

The neuroanatomy of the hippocampal region has suggested a distribution of various computational roles across its subfields. The main input sources is the entorhinal cortex (EC), classically divided into a medial and lateral part, shortened in MEC and LEC respectively. Notably, the former has been consistely linked with more spatial information, such as position-like signals from grid cells populations, while the latter with sensory and object-related features. The canonical route of entorhinal projections first targets the dentate gyrus (DG), with the role of reducing interference among similar neural patterns. From there, the following step is storing the preprocessed signal within the recurrent connections of the CA3 subfield neurons, set as an autoassociative network [13, 19]. Next, CA1 receives CA3 activity through the Schaffen collaterals but also direct entorhinal inputs through the temporoammonic pathway. Selective disruption of CA3 NMDA receptor function impairs recall when memories must be retrieved from partial cues, and CA3 and CA1 ensembles show distinct remapping profiles across changes in environmental context [9, 17]. More recent experiments and models have further developed accounts of recurrent CA3 dynamics and spatially structured associative representations [2, 11]. For the present framework, these findings permit CA3 activity to be treated as a retrieval cue, or key, that selects a stored episode. They do not, however, by themselves determine how the retrieved content is reformatted at hippocampal output so that downstream circuits can reliably read it out.

CA1 is not merely a passive relay: its representations can be formed or reorganized rapidly. Behavioral-timescale synaptic plasticity (BTSP) can induce a place field after a single dendritic plateau event and can bidirectionally reshape an existing field, with the direction and spatial distribution of plasticity depending on the relation between the plateau event and the pre-existing field [1, 15]. EC layer 3 activity can provide an instructive signal for learning-related changes in CA1 population activity [6]. Recent work further links BTSP-like dynamics to trial-by-trial field shifting, continued representational change, and the progressive formation of stable, task-relevant place codes [12, 21]. BTSP is therefore relevant not only to the formation of an isolated spatial representation, but also to the ongoing reorganization of CA1 codes as salient locations, cues, and task contingencies change. Computationally, BTSP-like dynamics can support one-shot content-addressable memory, demonstrating that this class of learning rule can in principle couple rapid storage with partial-cue retrieval [23]. Together, these findings are consistent with a functional division in which CA3 conveys retrieved associative content, whereas EC-related input contributes an instructive signal that gates or biases how the CA1 map is revised.

CA1 representations are also context sensitive. Changes in cue configuration can alter place-cell firing rates, field locations, or the active population, producing rate remapping and global remapping [10, 16]. In global remapping, the population code and/or spatial arrangement of fields changes substantially across contexts; in rate remapping, much of the spatial code is retained while in-field firing rates change with sensory or contextual variables. The entorhinal inputs received by CA1 provide a natural route through which spatial and nonspatial information may be combined. MEC populations are, on average, more strongly spatially modulated, whereas LEC populations often show weaker allocentric spatial signals in simple environments and can represent objects, object locations, and other salient sensory content [3, 7, 8]. This division is preferential rather than absolute; our use of separate “MEC-like” and “LEC-like” input partitions is therefore a modeling abstraction, not a claim of strict anatomical or functional segregation.

The model instantiates this prior structure through pretraining. Before a new episodic association is written, an EC–CA1–EC autoencoder learns a representational basis together with a corresponding decoder. This procedure is a computational device, not a proposal that CA1 is literally optimized by backpropagation during a separate developmental phase. Biologically, it abstracts the idea that development, accumulated prior experience, and relatively stable entorhinal–hippocampal connectivity furnish a structured repertoire on which rapid episodic learning can operate; related hippocampal models likewise use pre-existing spatial or representational scaffolds to organize new memories [2, 14, 22]. In deep learning, pretraining plays an analogous engineering role by establishing reusable features before adaptation to a new task, although the learning mechanisms and biological substrates differ substantially [24]. Freezing the decoder after pretraining makes the model's coordinate constraint explicit: newly learned CA1 states must remain expressible in a fixed representational system that a downstream readout can continue to interpret. It also prevents encoder–decoder co-adaptation from concealing changes in the geometry of the CA1 code.

These observations motivate a representational constraint. If rapid plasticity writes an arbitrary CA1 pattern while a downstream readout remains fixed, a pattern may retain information about an episode yet become functionally unusable because it is expressed in coordinates that the readout no longer interprets correctly. Work outside the hippocampus illustrates related constraints on population-level learning. In motor-cortex brain–computer-interface paradigms, animals rapidly learned perturbations that could be solved using activity within a pre-existing neural manifold, whereas perturbations requiring activity outside that manifold were more difficult to learn on the same timescale [18]. Short-term adaptation could also proceed through reassociation, in which an established repertoire of population activity patterns was linked to new behavioral outputs [5]. Across longer timescales, stable latent population dynamics supported reliable behavioral decoding despite turnover in the individual recorded neurons, although decoders defined directly on raw neural activity did not remain equally stable [4]. We use the narrower term *readout compatibility* to denote the requirement that a newly instructed CA1 pattern remain expressed in a representational coordinate system that a fixed output pathway can interpret. On this view, the relevant long-term object need not be a fixed CA1 activity vector; it can instead be a stable encoding–decoding relation within which individual CA1 representations are allowed to change.

We therefore study a deliberately minimal model rather than a detailed biophysical reconstruction. EC input is transformed into a discriminable CA3-like retrieval key. A pretrained and subsequently frozen EC–CA1–EC autoencoder defines both an instructed CA1 target and a fixed output decoder, while plastic CA3–CA1 synapses associate the retrieval key with that target. We compare a direct instructed-write rule, which writes the entorhinally specified CA1 target into the association, with a bounded error-driven rule, which corrects the discrepancy between instructed and recalled CA1 activity. This construction isolates four questions. First, does one-shot reconstruction depend on readout compatibility rather than information content alone? Second, what computational trade-off follows from correcting recall error rather than writing the instructed target directly? Third, can changes in cue arrangement produce a mixture of stable and cue-dependent CA1 tuning? Finally, how does recall degrade when the spatial or cue-like component of EC input is selectively corrupted? The resulting claims are computational and conditional. The model is intended as a constructive demonstration of a representational principle, not as evidence that biological CA3 or CA1 implements the exact architecture, pretraining procedure, decoder constraint, or synaptic update rules used here.

2 Results

2.1 A minimal retrieval-key, CA1-instruction architecture

The model contains four functional components (Fig. 1). EC input projects through a fixed sparse map to a CA3-like population, producing a low-activity retrieval key. A separately pretrained and frozen EC–CA1 encoder supplies an instructed CA1 target, denoted x_{IS}. Plastic CA3–CA1 weights associate the key with this target, while the corresponding pretrained CA1–EC decoder remains fixed. During recall, EC input regenerates a retrieval key, the learned CA3–CA1 weights reconstruct a CA1 state, and the fixed decoder maps that state back to EC output. The model therefore separates selection of associative content from its expression in the pretrained output basis.

We considered a direct instructed-write rule (labeled entorhinal-driven in Fig. 1B) and a bounded error-driven rule. The direct rule writes an outer-product association gated by the instructed CA1 target; it does not ask whether the current recalled state is already correct. We use it in the compatibility experiment because it exposes the coordinate constraint with minimal additional dynamics. The error-driven rule instead compares instructed and recalled CA1 activity, potentiating key synapses for underexpressed CA1 components and depressing them for overexpressed components. Consequently, repeated presentations correct a residual error rather than rewriting the full target. We use this rule for the cue-track and degradation experiments, where laps recur and cue assignments can change. Both rules are BTSP-inspired abstractions of rapid instructed plasticity rather than temporal models of plateau potentials.

A paired fixed-parameter control quantified the resulting trade-off (Supplementary Fig. S1). With intact input, the error-driven rule produced higher output–target cosine than the direct instructed-write rule (0.945 versus 0.848; paired difference 0.098, 95% confidence interval 0.090–0.105) and substantially higher nearest-target identity (0.673 versus 0.179; difference 0.494, 0.454–0.534). Under 90% whole-input degradation, however, the direct rule retained the higher output cosine (0.513 versus 0.393; difference 0.120, 0.093–0.146). Thus error correction sharpened clean-state discrimination under the shared parameterization, whereas direct writing produced a more graded similarity signal under severe corruption. This is a matched-parameter comparison, not evidence that either rule is globally optimal.

Track inputs combine a medial entorhinal cortex (MEC)-like spatial component and a lateral entorhinal cortex (LEC)-like cue component (Fig. 1D). Two cue identities occur at fixed positions and can exchange locations across blocks. For degradation tests, plasticity is frozen after clean storage, and a fixed fraction of either the MEC or LEC input units is set to zero throughout each probe lap (Fig. 1E).

2.2 A fixed decoder exposes a readout-compatibility constraint

We first isolated the relationship between the instructed CA1 code and the fixed output decoder (Fig. 2). Twenty independently initialized autoencoders were trained on sparse EC patterns and then frozen. For each seed, 28 novel patterns were stored once under five paired conditions. In the aligned condition, the instructed CA1 target remained in the representational coordinates learned by the decoder. In the fixed-permutation condition, the same non-identity permutation was applied to every instructed target while the decoder was left unchanged. In the matched-decoder condition, the corresponding decoder columns were permuted together with the target. Random-content and

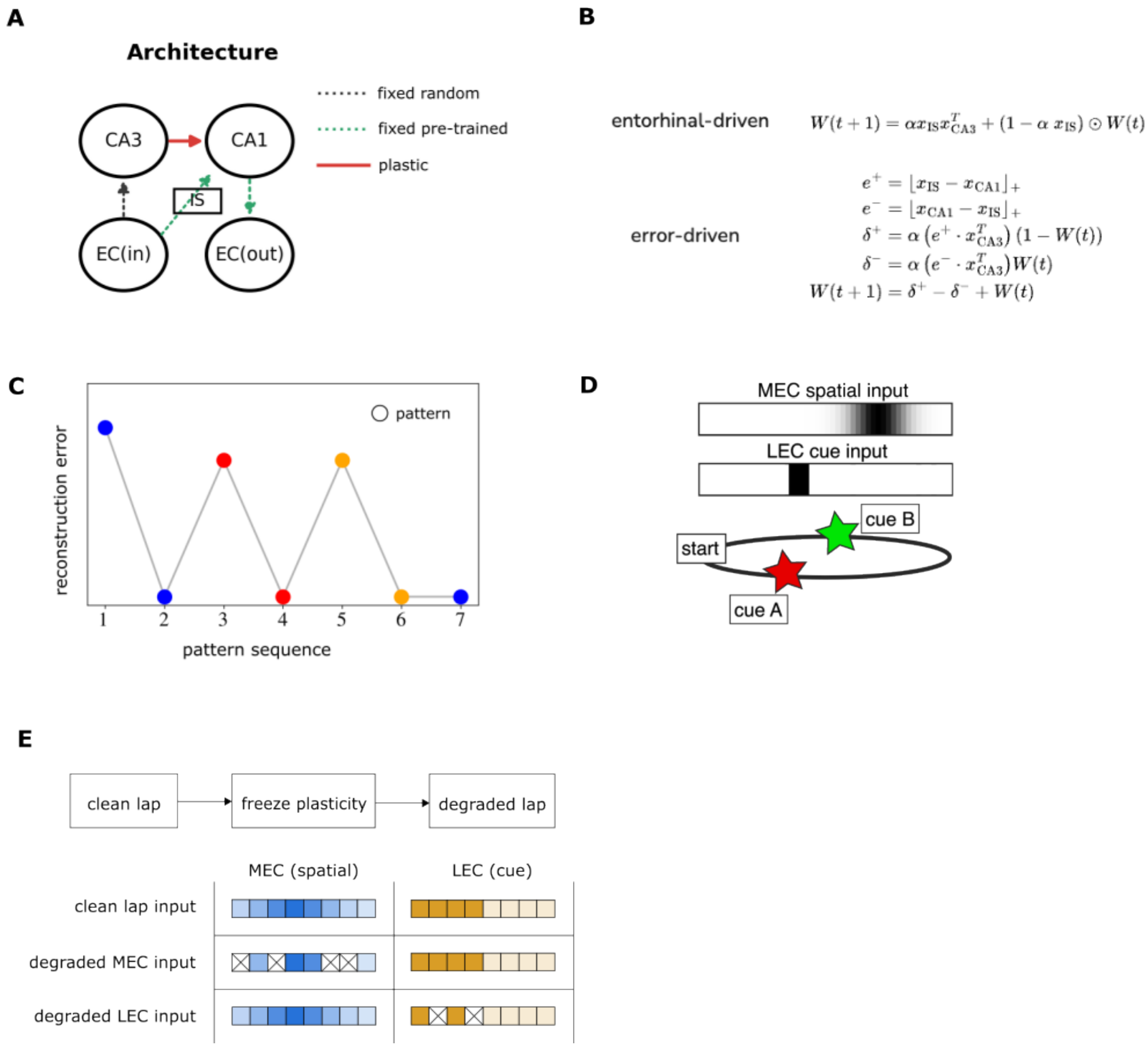

Figure 1: **Model architecture and simulation protocol.** **(A)** EC input produces a fixed sparse CA3-like retrieval key and a pretrained instructed CA1 target; plastic CA3–CA1 weights store their association and a fixed pretrained CA1–EC pathway decodes recall. **(B)** Direct instructed-write (entorhinal-driven) and bounded error-driven plasticity variants. **(C)** Schematic reconstruction error across first and repeated presentations of example patterns; colors denote pattern identity. **(D)** Cue-track input composed of an MEC-like spatial component and an LEC-like cue component. **(E)** Selective degradation protocol. Clean laps are stored before plasticity is frozen; a fraction of MEC or LEC units is then zeroed using a mask fixed within each probe lap and resampled across probes.

no-plasticity conditions controlled for an incorrect association and an unwritten memory, respectively.

Aligned instructions produced a mean output–target cosine of 0.804 (95% confidence interval, 0.788–0.820), whereas a fixed coordinate permutation reduced it to 0.136 (0.125–0.148; Fig. 2A). Every paired seed showed the same direction of change; the mean paired reduction was 0.668 (0.648–0.688). Permuting the decoder together with the instruction restored the mean cosine to 0.804. Random-content and no-plasticity controls remained low (0.182 and 0.195). The matched-decoder restoration is an algebraic coordinate control rather than an independent memory mechanism: the permutation preserves the target's information but changes its meaning to the unchanged decoder. Reconstruction therefore depends on compatibility between the instructed CA1 geometry and the fixed output basis, not on information content alone.

We next exposed the model to 40 track laps in which two cues exchanged positions every 10 laps. A paired no-swap control used the same seeds, pretrained encoder and decoder, CA3 wiring, MEC trajectories, lap count, and plasticity parameters, but retained cue assignment $[0, 1]$ throughout. After every lap, learning was paused and the model was probed in its currently scheduled context. Mean unit-wise tuning similarity to the preceding lap fell to 0.446 at cue-swap transitions (95% confidence interval, 0.423–0.468), compared with 0.831 at the corresponding transitions in the no-swap control (0.816–0.847; Fig. 2B). The paired difference was -0.385 (-0.409 to -0.361). At non-swap transitions in the alternating schedule, similarity remained 0.821 (0.808–0.834). Thus the abrupt changes were associated with the cue sequence rather than merely with continued learning across laps.

After learning, plasticity was frozen and the model was probed in the two cue arrangements. CA1 units showed heterogeneous spatial tuning across contexts (Fig. 2C,D). Some fields remained approximately stable, some changed amplitude in a rate-like manner, and others shifted their preferred position with cue identity. The seed-level mean of unit-wise, mean-centered tuning similarity was 0.175 (95% confidence interval, 0.155–0.194), and mean absolute cue modulation was 0.174 (0.171–0.177). These are descriptive summaries, not a quantitative fit to recorded CA1 populations or a formal classification of rate versus global remapping. Operationally, changing the cue arrangement changes both the EC-derived instructed target and the CA3-like retrieval key, producing a mixture of spatially stable and context-dependent recalled CA1 activity.

2.3 Selective EC degradation dissociates cue and spatial recall

We then tested whether the stored association remained usable when one EC input modality was selectively corrupted (Fig. 3). Models were trained on 20 intact laps with a fixed cue arrangement. Plasticity was frozen, and independent masks removed 0, 25, 50, 75, or 90% of eligible input units. Each condition used 12 masks per seed and 20 seeds. The principal comparison was between the normal sparse CA3-like retrieval map and a dense common map in which every unit received the same average over EC inputs. The latter produced identical pre-activation across units and deliberately removed item-specific key structure. It is therefore a degenerate-key control, not a sparsity-matched control.

With selective LEC degradation, the normal sparse retrieval map supported perfect cue identity at baseline and declined to 0.756 accuracy at 90% dropout (95% confidence interval, 0.636–0.877; Fig. 3A). The seed-level endpoint change was -0.244 (-0.364 to -0.123). Over the same range, MEC output–target cosine declined from 0.899 to 0.776, a change of -0.123 (-0.132 to -0.114; Fig. 3B), consistent with relative preservation of the intact spatial component. The dense common-key control

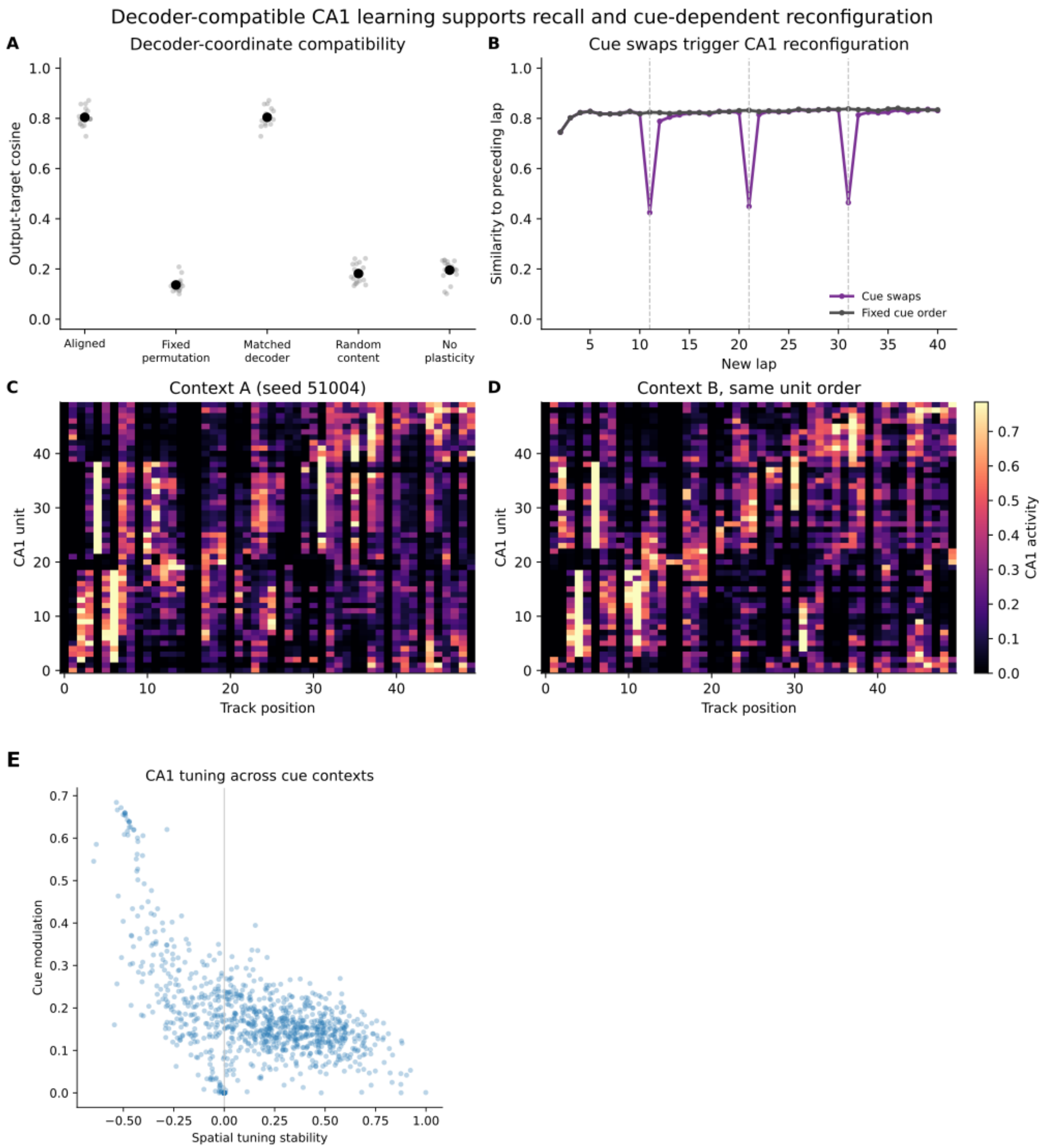

Figure 2: **Readout compatibility and cue-dependent CA1 reconfiguration. (A)** Output–target cosine across aligned, fixed-permutation, matched-decoder, random-content, and no-plasticity conditions. Grey points are seed means across stored memories; black points and error bars show mean ± SEM across 20 seeds. **(B)** Mean unit-wise tuning similarity between the scheduled-context probe after each lap and the corresponding probe after the preceding lap. Purple denotes the alternating cue schedule and grey the matched no-swap control; dashed lines mark cue exchanges. Curves show mean ± SEM across 20 paired seeds. **(C,D)** CA1 activity on the same track under the two cue arrangements, shown for the realization closest to the median population stability and modulation. Units have the same order in both heat maps and are sorted by preferred position across contexts. The heat maps illustrate stable, rate-like, and field-shifting responses but do not formally classify remapping type. Color limits are shared.

remained at chance cue accuracy (0.5) and low spatial-output cosine (approximately 0.30) even without degradation.

The complementary MEC degradation produced the opposite dissociation. With the normal sparse retrieval map, nearest-position accuracy declined from 0.655 at baseline to 0.074 at 90% dropout (chance = 0.02; endpoint change −0.581, 95% confidence interval −0.617 to −0.545; Fig. 3C). Cue identity remained perfect for every seed across the same degradation levels (Fig. 3D). The dense common-key control again remained at chance for position and cue identity. Thus, selective removal of cue-like or spatial input preferentially impaired the corresponding output measure while leaving the intact modality available. The result establishes partial-cue robustness in this architecture, but not recurrent pattern completion: recall degrades gracefully while retrieval keys remain discriminable, and the model does not recover content that becomes unidentifiable at extreme corruption.

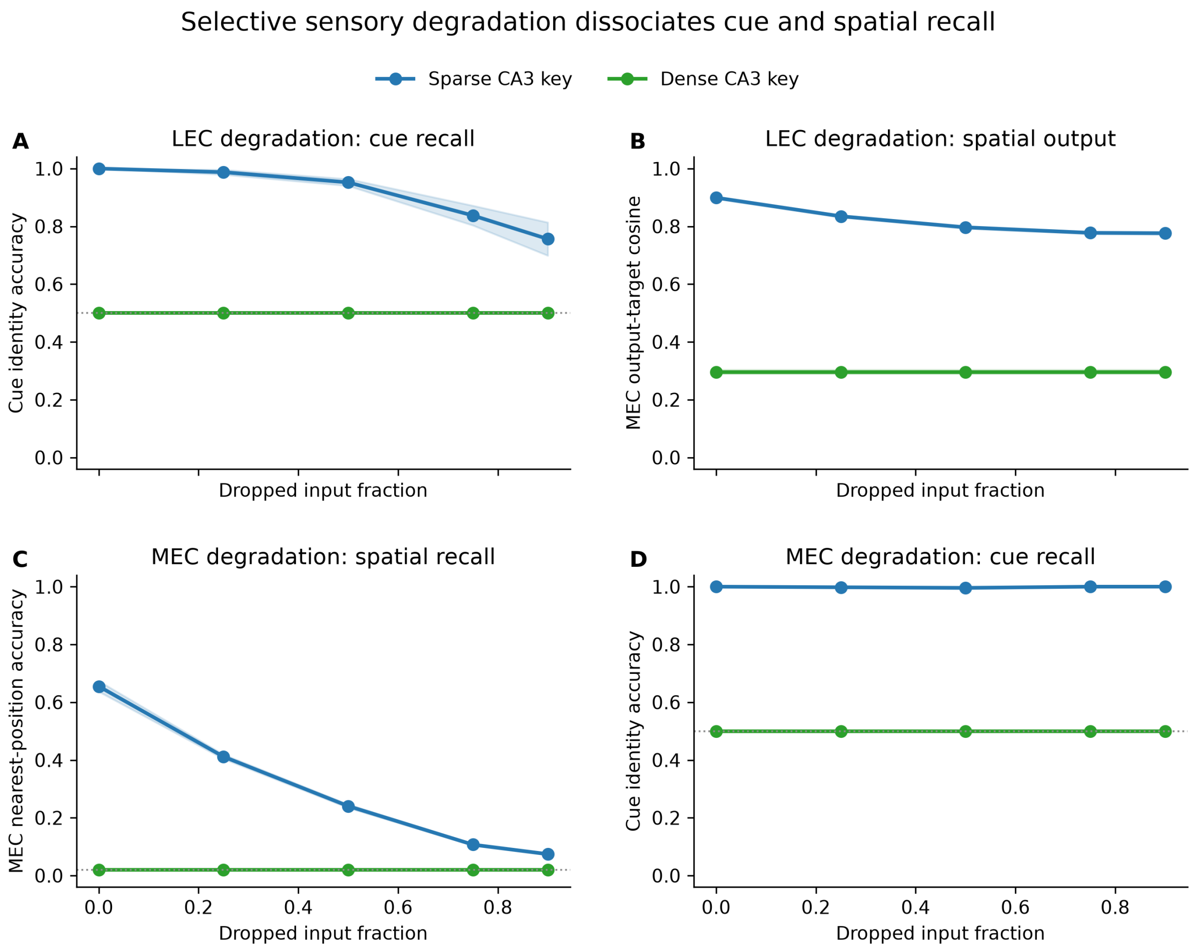

Figure 3: **Modality-specific recall under selective EC degradation.** **(A)** Cue identity accuracy during LEC-only degradation. **(B)** MEC output–target cosine during LEC-only degradation. **(C)** MEC nearest-position accuracy during MEC-only degradation. **(D)** Cue identity accuracy during MEC-only degradation. Curves show mean ± SEM across 20 seeds after averaging 12 masks within each seed. Blue denotes the normal sparse CA3-like retrieval map and green the dense common-key control, which collapses item-specific key structure and is not sparsity matched. Dotted horizontal lines denote chance accuracy where applicable. Plasticity was disabled throughout degraded recall.

3 Discussion

The central result is a representational constraint rather than a new memory-capacity claim. In this minimal architecture, a CA3-like retrieval key selects stored associative content, plastic CA3–CA1 weights re-express that content as a CA1 state, and a fixed decoder determines whether the state remains usable at the output. A coordinate permutation leaves the instructed target's information unchanged yet disrupts reconstruction when the decoder is not transformed with it. The matched-decoder rescue makes the logic exact: rapid plasticity must preserve, or co-adapt with, the representational coordinates assumed by subsequent stages. This motivates viewing CA1 not only as a locus where individual experiences are stored, but as an adaptive interface that reformats retrieved content within a structured neural space. Such a space could allow new experiences to remain decodable by established recipient circuits even while participating neurons and fields change.

This proposal complements, rather than replaces, existing accounts of hippocampal memory. Autoassociative CA3 theories explain how a stored state can be selected from a partial cue, while BTSP models explain how strong, temporally broad instructive events can rapidly alter CA1 synapses [1, 17, 23]. The present model abstracts the first operation as a non-recurrent CA3-like key and asks what must additionally hold when selected content is expressed at hippocampal output. The relationship to neural-manifold work is conceptual: those studies concern motor cortical learning, not hippocampal memory, but similarly demonstrate that an unchanged readout assigns meaning only to particular population coordinates [5, 18]. Readout compatibility is therefore a general systems constraint instantiated here in a hippocampal circuit motif. The present simulations do not test retention across days or years; "long-term decodability" is a hypothesis about what preserving this representational relation could enable, not a measured outcome.

Pretraining and rapid plasticity play distinct computational roles in that proposal. Pretraining establishes a sparse representational basis together with the decoder that gives its coordinates meaning; rapid CA3–CA1 learning expresses a new, key-addressable experience within that basis. This resembles feature reuse after pretraining in deep networks [24], but the analogy is functional rather than mechanistic. In the biological circuit, the basis could reflect development, prior learning, and ongoing entorhinal–hippocampal organization rather than a separate optimization phase. The fixed decoder is an assay of compatibility, not a claim that biological recipient pathways are immutable. The important prediction is that changes in CA1 codes and their recipient readouts must remain coordinated.

The cue-track simulations connect this constraint to context-dependent reconfiguration. EC-driven changes in both the instructed target and CA3-like retrieval key generated heterogeneous CA1 fields when cue identities exchanged positions. A matched no-swap control remained stable at the corresponding lap boundaries, tying the abrupt reconfiguration to the changed cue sequence rather than to continued training alone. Some responses were spatially stable, some changed amplitude in a rate-like manner, and others shifted preferred position. This behavior is qualitatively consistent with cue-sensitive rate and field remapping and with evidence that EC-directed BTSP forms, shifts, and stabilizes CA1 fields around behaviorally relevant features [6, 10, 12, 21]. On this interpretation, BTSP is an update operation on a spatial cognitive map: it can rapidly insert a field, weaken or move an existing field, and progressively allocate representation to task-relevant structure. Our bounded error-driven rule captures only the computational motif of bidirectional, target-dependent correction. It is not a temporal or molecular model of BTSP, nor is the simulation a quantitative model of biological remapping. Without comparison to recorded field statistics or a validated classifier, the present results should not be labeled as formal global or rate remapping. Their narrower prediction is that context-dependent reconfiguration should be evaluated not only by how much CA1 activity changes, but also by whether the changed population code remains decodable by its recipient circuit.

The two learning rules clarify why the form of this update matters. The direct instructed-write rule stores the target association regardless of current recall, whereas the bounded error-driven rule changes only components that are under- or overexpressed. Under matched parameters, error correction greatly improved clean nearest-target discrimination but lost similarity faster under severe input degradation. This selectivity–robustness trade-off is computationally useful for interpreting the main experiments: the error-driven rule was chosen for repeated, changing cue sequences because it can correct residual and interfering activity, not because it is uniformly more noise tolerant. A full comparison would optimize each rule separately and vary memory load, cue overlap, and learning rate.

The degradation experiments distinguish partial-cue robustness from mechanistic pattern completion. Recall remained above chance under substantial input loss, an operational signature often used to describe completion. Yet the CA3-like population in this model has no recurrence or attractor settling; robustness arises from distributed input-to-key mappings and learned key-to-content associations. The result therefore shows that degraded-cue recall does not by itself identify a recurrent completion mechanism. A biological claim about CA3 pattern completion would require explicit recurrent dynamics and comparison with interventions that alter those dynamics [11, 17].

The control comparison also has a deliberately limited interpretation. The dense common map forces all CA3-like units to receive the same EC average, so it removes retrieval-key discriminability while also changing sparsity. Its failure establishes that a non-specific common key cannot support these associations; it does not establish that sparse wiring is uniquely necessary or optimal. Likewise, the MEC/LEC degradation dissociation is partly expected from the constructed input and readout partitions. Its value is as an internal validation that a shared CA1 representational basis can preserve access to an intact component, not as a discovery of anatomical modularity. Distinguishing the contribution of sparsity from dimensionality, key separation, and connection density will require matched controls.

Several limitations further define the scope of the proposal. The model omits dentate gyrus, recurrent CA3 dynamics, attractor competition, temporally extended sequence mechanisms, inhibition, consolidation, and any explicit long-term drift of neurons or readouts. MEC and LEC are represented as simplified input halves despite mixed selectivity in both regions. The pretrained CA1 representational basis is obtained from a shallow autoencoder rather than a biologically specified developmental or learning process, and the plasticity equations are bounded abstractions rather than temporal or molecular models of BTSP. The compatibility manipulation is deliberately synthetic, and its matched-decoder rescue is an equivariance control rather than independent biological evidence. Finally, only two cue identities and one memory load were used in the main track experiments; capacity, interference, and multi-context scaling remain unresolved.

These limitations suggest concrete tests. In the model, performance should depend more strongly on compatibility between an instructed CA1 state and the fixed output basis than on the magnitude of CA1 change alone. Partial-cue recall should track retrieval-key discriminability across sparsity-matched control maps, and increasing the number of cue-pair sequences should eventually expose an interference-dependent capacity limit. Experimentally, the broad prediction is that learning-related CA1 reconfiguration can remain behaviorally useful when recipient populations preserve or rapidly acquire an appropriate readout, whereas equally large but readout-incompatible changes should impair retrieval.

Within these bounds, the model offers a compact proposal linking a previously established representational basis, rapid instructed plasticity, retrieval-key-based selection, and downstream readability. CA1 activity can change substantially as a cognitive map is updated, yet remain useful if retrieved content is reformatted into coordinates that recipient circuits can interpret. Readout compatibility is therefore a useful, testable axis for evaluating models of rapid hippocampal learning and for asking how a changing hippocampal code could support decodable experience over longer timescales.

4 Methods

4.1 Implementation, reproducibility, and statistical treatment

Simulations were implemented in Python using NumPy and PyTorch and run on CPU. All reported conditions used the same fixed set of 20 root seeds (51001–51020); conditions within an experiment were paired by root seed. Each experiment writes an immutable artifact containing the resolved configuration, compressed arrays, tidy source data, a report, the Git revision, and a scientific digest. Figure scripts read these artifacts without rerunning simulations.

The unit of computational replication was the root seed, not an individual memory, track position, unit, or corruption mask. Values were first averaged over these nested observations within each seed and then summarized across seeds. Curves and error bars show mean ± SEM. Confidence intervals reported in the text are two-sided 95% Student-t intervals over the 20 seed-level values; for contrasts they were calculated from paired seed-level differences. These intervals quantify sensitivity to the sampled initializations and random inputs, not uncertainty over a biological population. No null-hypothesis significance tests or multiple-comparison procedures were used, and all analyses should be interpreted as descriptive simulation results.

4.2 EC patterns and the pretrained CA1 representational basis

All populations had dimension 50. A one-hidden-layer autoencoder mapped EC input to a 50-unit CA1 representation and back to EC output. CA1 activity used a differentiable top-K sparse activation with $K_{\mathrm{CA1}} = 5$; output activity used a sigmoid decoder. The encoder and decoder were trained by Adam to minimize mean-squared reconstruction error. Biases were disabled. This pretraining occurred before CA3–CA1 episodic storage; the resulting encoder supplied all instructed targets and both the encoder and decoder were then held fixed. Pretraining should therefore be understood as a way to define a shared representational basis and its readout, not as a simulated biological learning phase. For the readout-compatibility experiment, the autoencoder was trained for 1024 epochs on 2048 unique binary patterns with five active units and validated on 256 held-out patterns. For track experiments, it was trained for 256 epochs on 1000 track samples and validated on 250 independently generated samples. Mean validation reconstruction cosine across seeds was 0.99998 for the compatibility experiment, 0.99946 for cue remapping, and 0.99736 for the degradation experiments.

Track inputs contained 25 MEC-like and 25 LEC-like units. MEC activity varied with position on a circular track. Two fixed LEC cue patterns were presented at positions 10 and 30 on a 50-position lap. Cue and spatial widths, gains, and binarization settings are recorded in the resolved configuration files distributed with the result artifacts.

4.3 CA3-like retrieval keys and CA3–CA1 plasticity

EC input was projected to 50 CA3-like units through a fixed sparse matrix followed by a differentiable top-K activation. The compatibility experiment used two EC inputs per CA3-like unit, $K_{\mathrm{CA3}} = 5$, inverse-temperature parameter $\beta_{\mathrm{CA3}} = 200$, and learning rate $\alpha = 0.08$. Track experiments used 29 EC inputs per CA3-like unit, $K_{\mathrm{CA3}} = 3$, $\beta_{\mathrm{CA3}} = 170.6$, and $\alpha = 0.092$.

For the direct instructed-write rule (called entorhinal-driven in the implementation), CA3–CA1 weights were updated sequentially as

$$W_{t+1} = (1 - \alpha x_{\mathrm{IS}}) \odot W_t + \alpha x_{\mathrm{IS}} x_{\mathrm{CA3}}^{\mathsf{T}}, \tag{1}$$

where broadcasting applies the instructed CA1 target to each row of W. For the bounded error-driven rule,

$$\begin{aligned} e^{+} &= [x_{\mathrm{IS}} - x_{\mathrm{CA1}}]_{+}, & e^{-} &= [x_{\mathrm{CA1}} - x_{\mathrm{IS}}]_{+}, && (2)\\ \delta^{+} &= \alpha(e^{+} x_{\mathrm{CA3}}^{\mathsf{T}})(1 - W_t), & \delta^{-} &= \alpha(e^{-} x_{\mathrm{CA3}}^{\mathsf{T}})W_t, && (3)\\ W_{t+1} &= \mathrm{clip}_{[0,1]}(W_t + \delta^{+} - \delta^{-}). &&&& (4) \end{aligned}$$

The instructed target x_{IS} was the sparse output of the pretrained EC–CA1 encoder. Recalled CA1 activity was computed from $W x_{\mathrm{CA3}}$ using the same target sparsity, then passed through the fixed pretrained decoder. The direct instructed-write update depends on the target and presynaptic key but not on the current recall error. The bounded error-driven update separates the residual into positive and negative components, so correctly expressed CA1 components receive no direct update while under- and overexpressed components drive potentiation and depression, respectively.

4.4 Plasticity-rule control

The supplementary plasticity control used the same autoencoder, CA3 wiring, track inputs, 20 root seeds, 20 clean training laps, learning rate, and all other memory parameters for both rules. Only the plasticity-rule selector differed. After clean storage, plasticity was frozen and 0, 25, 50, 75, or 90% of the full EC input was removed using 12 independently sampled masks per fraction and seed. Output–target cosine and nearest-target identity were first averaged across masks and track positions within each seed and then summarized across seeds. Because parameters were inherited from the error-driven track configuration and were not optimized separately for the direct instructed-write rule, this experiment tests behavior under a shared parameterization rather than the best attainable performance of either rule.

4.5 Readout-compatibility experiment

For each seed, 28 held-out sparse EC patterns were assigned CA3-like retrieval keys and pretrained CA1 targets. Memories were written once in a seed-specific random order. The aligned condition used the original target and decoder. The fixed-permutation condition applied one non-identity CA1 permutation to all targets while leaving the decoder fixed. The matched-decoder condition used the same permuted targets and correspondingly permuted decoder columns. The random-content condition associated each key with another memory's target using a derangement. The no-plasticity condition retained zero CA3–CA1 weights. Recall was evaluated with output–target cosine, reconstruction error, top-K overlap, and nearest-target identity.

4.6 Cue-reconfiguration experiment

Models learned 40 complete laps. Cue identities exchanged positions every 10 laps, alternating between assignments $[0, 1]$ and $[1, 0]$. Plasticity was then frozen and one lap from each context was recalled. For each CA1 unit, spatial stability was the Pearson-style cosine between mean-centered context tuning curves. Cue modulation was the mean absolute activity difference between contexts. These two summaries quantify context sensitivity but do not by themselves classify a unit or population as undergoing rate or global remapping. Heat maps show the seed whose mean stability and modulation were closest to the across-seed medians; units were ordered by preferred position across the two contexts.

The temporal control paired this alternating schedule with a no-swap schedule that retained assignment $[0, 1]$ for all 40 laps. Within each seed, the two conditions shared the pretrained autoencoder, initial CA3–CA1 weights, CA3 wiring, MEC input trajectory, lap count, and plasticity parameters. Only the LEC cue assignment differed after the first 10 laps. After each training lap, plasticity was paused and both fixed context probes were evaluated; the probe matching the context scheduled for that lap was retained. Consecutive-lap tuning similarity was calculated for each CA1 unit after mean-centering its activity across track positions and then averaged across units. Cue-swap transitions were compared with the same lap transitions in the paired no-swap model. Intervals summarize seed-level means across 20 paired root seeds.

4.7 Selective degradation experiments

Models stored 20 clean laps with cue assignment $[0, 1]$. During frozen recall, a fixed random subset of eligible EC units was zeroed throughout the complete lap. Masks were resampled independently between probes. Dropout fractions were 0, 0.25, 0.5, 0.75, and 0.9, with 12 masks per fraction and seed. LEC degradation operated only on the final 25 input units; MEC degradation operated only on the first 25.

Cue identity was decoded at the two cue positions by cosine comparison of recalled LEC output with the two fixed cue templates. Spatial accuracy was the fraction of recalled MEC outputs whose nearest clean MEC target corresponded to the correct track position. MEC output cosine was averaged over all positions. The dense common-key control replaced the EC–CA3 map by a matrix with every entry equal to $1/50$, causing all CA3-like units to receive the same input average. Because this manipulation collapses item-specific pre-activation and is not matched to the normal map for sparsity, it tests the requirement for a discriminable retrieval key rather than the isolated effect of sparse connectivity.

Data and code availability

Code, resolved configurations, source-data tables, and immutable result artifacts are included with this repository. A permanent public archive and accession identifier will be added before submission.

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To be completed.

Author contributions

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Supplementary material

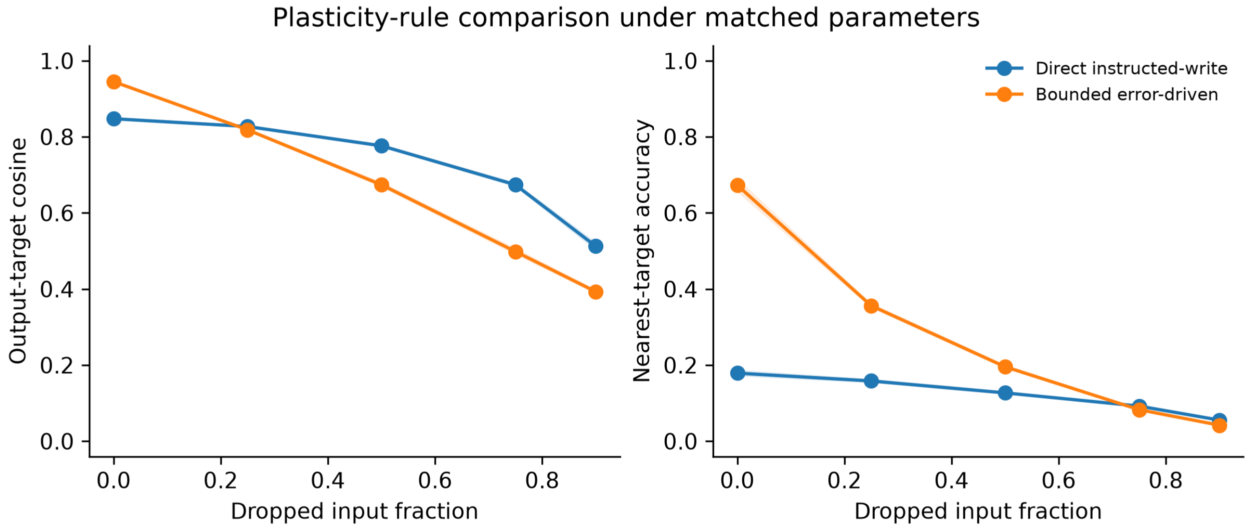

Figure S1: **Matched-parameter comparison of the two plasticity rules.** Output–target cosine (left) and nearest-target identity accuracy (right) after 20 clean training laps, followed by frozen recall with increasing fractions of the full EC input removed. Curves show mean ± SEM across 20 root seeds after averaging 12 masks within each seed. All parameters, inputs, pretrained autoencoders, and CA3-like retrieval maps were shared across the paired rule conditions; only the update equation differed. The error-driven rule improved clean-state target discrimination, whereas the direct instructed-write rule retained higher output similarity under severe corruption. Parameters were not optimized separately for each rule.